

# Rishabh Shah

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## Education

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| <b>Rutgers University - New Brunswick</b><br>Master of Science (MS) in Computer Science, <b>GPA: 3.83 / 4.0</b>                                                 | Sep. 2025 - May. 2027 |
| <b>University of Mumbai</b><br>Bachelor of Technology in Artificial Intelligence & Data Science, <b>Honors</b> in Computational Biology, <b>GPA: 3.25 / 4.0</b> | Dec. 2021 - May. 2025 |

## Technical Skills

- **Programming & Scripting:** Python, C++, Java, C, SQL, Typescript
- **Full-Stack Development:** Next.js, Flask, Django, FastAPI, REST APIs, Microservices, JWT Authentication
- **Databases & Data Handling:** MySQL, PostgreSQL, MongoDB, Schema Design, Query Optimization, Data Modeling
- **Cloud & DevOps:** AWS (EC2, S3, Lambda), Docker, Azure, GCP, GitHub Actions, Git, Linux & Unix, System Monitoring
- **Additional Technologies:** Numpy, Pandas, Scikit-learn, TensorFlow, Keras, PyTorch, YOLO, OpenCV, OCR & ANPR

## Experience

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| <b>Aaditya Technologies</b> , Mumbai, India<br><i>Software Engineering Intern</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Jan. 2025 - Jul. 2025 |
| <ul style="list-style-type: none"><li>• Engineered and implemented backend services using Python, Next.js, and MySQL, designing modular and reusable APIs, refining relational database schemas, and optimizing complex SQL queries to improve data handling efficiency and reduce average response time by 22%</li><li>• Rolled out secure and scalable REST APIs with JWT-based authentication and role-based access control, enabling seamless inter-module communication, integrating structured logging, and enhancing data reliability by 32% while reducing integration and runtime issues</li><li>• Containerized and deployed scalable microservices using Docker and AWS (EC2, ECS), automating CI/CD workflows with GitHub Actions for build, test, and deployment pipelines, improving deployment speed by 45%, ensuring environment consistency, and minimizing downtime</li><li>• Collaborated across the software development lifecycle by designing, testing, and integrating production-ready features, following structured development workflows, adhering to version control best practices with Git &amp; GitHub, and supporting releases in cloud-based environments</li></ul> |                       |
| <b>Creative Line</b> , Mumbai, India<br><i>Web Developer Intern</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Jun. 2024 - Sep. 2024 |
| <ul style="list-style-type: none"><li>• Optimized a core Next.js app, reducing page load time by 18%, enhancing responsiveness via code-splitting, asset compression, and caching</li><li>• Automated key data flows by building and testing Next.js API routes, implementing error handling, enabling reliable frontend to backend integration, reducing manual operations by 38%, and accelerating data retrieval by 23% through optimized MySQL queries and indexing</li><li>• Conducted systematic testing, debugging, and code documentation, reproducing and resolving edge-case issues, improving overall application reliability, reducing development issues by 15%, and creating reusable documentation to improve onboarding and team workflows</li><li>• Maintained and enhanced reusable frontend components and backend API logic, adhering to clean architecture and modular design principles, refactoring legacy code where needed, improving maintainability, and supporting scalable feature development across the application</li></ul>                                                                                                                                         |                       |

## Technical Projects

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| <b>Traffic Violation Detection &amp; Automated Ticketing System</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | github.com/RishiiShah/Traffic |
| <ul style="list-style-type: none"><li>• Developed a YOLO-based computer vision pipeline detecting five traffic violations (speeding, red-light jumping, helmet-less riding, expired PUC/insurance) with 93.76% test accuracy, combining vehicle tracking, speed estimation, and rule-based logic to reduce false positives by 18%</li><li>• Engineered backend automation for challan generation and delivery using AWS S3 for storing evidence-backed PDF reports and Twilio API for SMS notifications, orchestrating end-to-end violation workflows reliably at scale, secure, and reducing manual enforcement effort by 90%</li><li>• Integrated OCR-based ANPR for license-plate recognition using vision inference, implementing frame selection, regex-based validation for Indian number plate formats, and latency-aware preprocessing (tripwire logic, frame handling) to achieve 30% faster inference</li></ul> |                               |
| <b>JARVIS: Voice Assistant with Smart Home Automation</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | github.com/RishiiShah/Jarvis  |
| <ul style="list-style-type: none"><li>• Developed a full-stack smart home automation system with voice and web control using Django, Azure IoT Hub, and Raspberry Pi, enabling secure real-time control of household appliances through cloud-to-device messaging and GPIO-based hardware actuation</li><li>• Integrated Llama-3 (70B) via Groq API for natural language understanding, implementing asynchronous request handling, prompt filtering, and multi-threading to achieve sub-300 ms response latency and 40% higher throughput under concurrent user queries</li><li>• Designed REST APIs for secure device communication and state management, ensuring 99.9% consistency between cloud and local device states, real-time synchronization, and fault-tolerant control of connected smart devices</li></ul>                                                                                                  |                               |
| <b>Music Genre Detection (ML Pipeline)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | github.com/RishiiShah/Music   |
| <ul style="list-style-type: none"><li>• Implemented a Bi-LSTM-based audio classification model using Librosa for MFCC, Chroma, and Mel features, achieving 98.73% accuracy</li><li>• Streamlined data preprocessing and training pipelines in Python, improving training efficiency by 22% and reducing memory usage by 15%</li><li>• Deployed quantized model as a lightweight inference API for real-time genre tagging with low compute overhead</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                               |

## Publications

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| <b>Intelligent Traffic Surveillance: A Vision-Based System for Detecting Traffic Rule Violations</b>                     | Sep. 2025 |
| Developed a vision-based system using YOLOv12 and OCR to automate traffic violation detection, achieving 93.76% accuracy |           |
| <b>Bridging Financial Data Gaps with WGAN-GP Generating Synthetic Time Series for Robust Models</b>                      | Aug. 2025 |
| Synthesized realistic financial time series using WGAN-GP, improving predictive model accuracy by 0.9%                   |           |
| <b>JARVIS: Voice Assistant with Smart Home Automation</b>                                                                | Jun. 2025 |
| Designed and deployed a voice-controlled IoT automation system using LLMs and Raspberry Pi to simplify daily routines    |           |